

SANYAM VERMA

@ sanyamv.iitd@gmail.com

+91 8287393623

in linkedin.com/in/sanyam-verma

https://github.com/SanyamVb

EXPERIENCE

Bayers | AI Engineer

Bangalore [July 2025 – Present]

- Architected the backend for **Bayer Employee Assistant** (enterprise internal platform) using a scalable **FastAPI** system serving **500+** users with **<100ms** average API latency and **99.9% uptime**
- Implemented **enterprise-grade authentication** using **Azure Active Directory** (Azure AD SSO), **OAuth2**, **JWT**, and **JWKS**, handling **1,000+** daily auth events with secure token validation
- Developed **23 RESTful APIs** using **FastAPI + Pydantic** with strict request/response validation, strong type safety, and clean API design
- Integrated **Microsoft Graph APIs** for enterprise automation, processing **500+** daily operations across **Outlook**, **Calendar**, **Teams**, and **User services**, and developed a **LiveKit-based voice agent** for real-time employee interactions
- Engineered **real-time session infrastructure** using **LiveKit**, **WebRTC**, and async event handling, supporting **50+** concurrent connections with **<300ms end-to-end latency**
- Maintained **4,000+** lines of production **Python** using **type hints (typing)**, **structured logging**, and **centralized exception handling framework**, improving system reliability and debuggability

Wipro Enterprises | Machine Learning Intern

Bangalore [May 2024 – July 2024]

Awarded Pre-Placement Offer (PPO) for exceptional performance and delivering scalable results in the key projects

- Automated & optimized** finalization of the Cushioning parameter by extensive data manufacturing and **custom-built model**
- Leveraged advanced **Data Engineering** techniques, **XGBoost**, self-modified **Monte-Carlo** method, **Bayesian techniques**
- Pioneered 2 Python based **GUIs** using **Streamlit**, attained **91%** accuracy while predicting final Cushioning parameters
- 720x** increased efficiency for cushioning design parameters finalization; project implemented in firm's workflow

PROJECTS

MultiModal chatbot | LlamaIndex | RAG [Jun'25]

- Engineered a **MultiModal RAG** based chatbot using **llamaindex**, capable of retrieving and responding from multimodal files.
- Enabled **long-term memory** and **reduced hallucination** through context-aware retrieval and node-level filtering
- Integrated **PostgreSQL + PGVector** for scalable vector storage, **Streamlit** for real-time UI, and **FastAPI** ensuring minimal latency.

Emotion-Based Music Recommender | ResNet50V2 | OpenCV [Jun'24]

- Achieved **71.32% accuracy** on facial emotion recognition using fine-tuned **ResNet50V2** and **TensorFlow** on FER2013
- Developed real-time emotion detection with **OpenCV** and **Haar Cascade**, integrating song suggestions based on emotions
- Compared and improved **modified CNN** and **ResNet50V2** for emotion classification using data augmentation and transfer learning

EDUCATION

B.Tech in Engineering and Computational Mechanics

Indian Institute of Technology Delhi

2021 – 2025

Relevant Coursework: Machine Learning in Mechanics, DeepLearning.AI[Knowledge Graphs for AI Agents(Neo4j)], Building Voice Agents using Livekit], Calculus, Linear Algebra, CampusX[NLP, Deep learning]

ACHIEVEMENTS

- Letter of Recommendation:** Endorsed by LoR from **Vice President & Head**, R&D, Wipro Enterprises.
- Best BTP award:** Mrs. Malti Singh Memorial undergraduate award for **Best B.Tech project**
- Pro5.ai Certified:** Ranked in the top 5% of professionals in Machine Learning by Pro5 certification
- GATE 2025:** Qualified Graduate Aptitude Test in Engineering 2025 - Data Science and Artificial Intelligence
- JEE Advanced 2021:** Secured **AIR-792 (OB)** in JEE-Advanced out of **1.4 lakh** aspirants across India.
- JEE Main 2021:** Secured **99.3 percentile** in JEE-Main, competing among **10.2 lakh** aspirants across India.

P.O.Rs

- Central Team Member | IIT Delhi [Apr'23 – May'24]:** Honored by *Significant Contribution award*.
- Department Head | Infinity Hyperloop [Dec'22 – Oct'23]:** Secured wins at *EHW-2023 & CHC-2023*

TECHNICAL SKILLS

Programming: C++, Python, TypeScript, MySQL, PostgreSQL, OpenCV, TensorFlow, PyTorch, NumPy, Pandas, SciPy, Scikit-learn, Matplotlib, Seaborn, Git, GitHub, GitHub Actions, REST APIs, GraphQL

Frameworks: LangGraph, LlamaIndex, LangChain, LiveKit, Streamlit, FastAPI, MLflow, Neo4j, Airflow, Docker, AWS, SQLAlchemy, Alembic, Pydantic, Redis based Langcache, WebRTC